ADDITIONAL FILE 2: SUPPLEMENTAL TABLES:

Table S.1. Sample size of WHI, HRS, and HANDLS datasets by exposure variables

	WHI [a] Training	WHI [a] Test	WHI [a] Total	HRS [b] Validation 1	HANDLS [c] Validation 2
Body mass index:					
Normal	34012	14751	48763	857	241
Overweight	33024	14212	47236	955	284
Obesity I	16655	7088	23743	631	223
Obesity II	6454	2712	9166	275	152
Obesity III	3318	1394	4712	167	137
Total	**93463**	**41057**	**133620**	**2885**	**1037**
Body mass index-specific waist circumference categories:					
≥ 80 cm (normal weight)	7076	3044	10120	318	129
≥ 90 cm (overweight)	8802	3711	12513	539	204
≥ 105 cm (obesity I)	2052	946	2998	298	92
≥ 115 cm (obesity II)	826	332	1158	657	55
≥ 115 cm (obesity III)	1662	693	2355	376	115
< 80 cm (normal weight)	26936	11707	38643	255	112
< 90 cm (overweight)	24222	10501	34723	172	80
< 105 cm (obesity I)	14603	6142	20745	103	131
< 115 cm (obesity II)	5628	2380	8008	48	97
< 115 cm (obesity III)	1656	701	2357	119	22
Total	**93463**	**41057**	**133620**	**2885**	**1037**

Abbreviations: BMI = Body mass index; HANDLS = Healthy Aging in Neighborhoods of Diversity across the Life Span; HRS = Health and Retirement Study; WC = Waist circumference; WHI = Women's Health Initiative.

Note: Body mass index (BMI) was categorized as normal weight (18.5 to <25 kg/m^2), overweight (25 to <30 kg/m^2), obesity class I (30 to <35 kg/m^2), obesity class II (35 to <40 kg/m^2), and obesity class III (≥40 kg/m^2).

[a] WHI sample sizes are based on postmenopausal women aged 50–79 years at enrollment (1993–1998) who were free of type 2 diabetes, had complete baseline data, BMI ≥18.5 kg/m², and >12 months of follow-up. The analytic sample was randomly split into 70% training and 30% test sets for model development and evaluation. BMI and WC were measured at baseline using standardized protocols, and joint BMI–WC categories were constructed using predefined thresholds.

[b] HRS sample sizes are based on women aged ≥ 50 years in the 2006 wave who were free of type 2 diabetes at baseline, had complete data on BMI and WC, BMI ≥ 18.5 kg/m², and were followed through 2020 (n=2,885). BMI categories were defined using standard cut points, and WC, measured during the Enhanced Face-to-Face assessment, was classified using BMI-specific thresholds. Joint BMI–WC categories were constructed by cross-classifying these measures to define mutually exclusive adiposity phenotypes. Counts reflect the analytic sample after all exclusions and harmonization with WHI definitions.

[c] HANDLS sample sizes are based on women aged 30—64 at baseline who were free of type 2 diabetes at baseline, had complete data on BMI and WC, BMI ≥ 18.5 kg/m², and were followed from 2004 until the latest available follow-up, for a maximum follow-up time of ~20 years, and a mean follow-up time of ~8 years (n=1,037). BMI categories were defined using standard cut points, and WC was classified using BMI-specific thresholds. Joint BMI–WC categories were constructed by cross-classifying these measures to define mutually exclusive adiposity phenotypes. Counts reflect the analytic sample after all exclusions and harmonization with WHI definitions.

Table S.2. Descriptive characteristics of HRS dataset by exposure variables [a]

Variable	Category	Overall (n=2885)	Normal (n=857)	Overweight (n=955)	Obesity I (n=631)	Obesity II (n=275)	Obesity III (n=167)	P-value
Age (years)	Mean ± SE	67.68 ± 0.40	70.03 ± 0.35	68.20 ± 0.35	66.16 ± 0.42	64.84 ± 0.55	63.12 ± 0.63	
Age group	50–59	25.91	20.18	24.37	27.22	36.1	42.47	<0.0001
	60–69	35.04	32.27	33.91	40.27	35.43	35.36	
	70+	39.05	47.56	41.72	32.51	28.47	22.17	
Race/ethnicity	Non-Hispanic White	82.73	87.82	82.72	77.85	79.85	79.78	<0.0001
	Non-Hispanic African American	7.51	5.55	4.82	10.3	11.92	15.13	
	Hispanic	7.65	4.24	10.02	10.41	5.24	5.08	
	Non-Hispanic Other	2.12	2.38	2.45	1.44	2.99	0	
Marital status	Never married	3.05	2.17	3.19	3.37	4.07	3.81	0.12
	Married/partnered	58.99	57.17	62.06	57.77	56.76	58.98	
	Separated/divorced	16.27	15.19	13.77	19.87	17.75	20.11	
	Widowed	21.69	25.47	20.98	18.98	21.43	17.1	
Education	Less than high school	16.46	14.08	18.41	18.18	16.27	11.4	0.088
	GED	4.18	2.87	4.41	4.24	5.04	7.91	
	High school graduate	33.44	33.32	33.31	33.75	34.48	31.97	
	Some college	24.51	22.9	24.17	25.13	25.09	31.44	
	College and above	21.4	26.82	19.7	18.7	19.11	17.29	
Household wealth	<$25,000	31.16	33.48	28.39	32.08	30.72	32.36	0.18
	$25,000–$124,999	57.81	54.03	58.95	58.25	60.6	64.35	
	$125,000–$299,999	9.07	9.37	10.56	8.2	8.5	3.28	
	$300,000–$649,999	1.55	2.41	1.86	0.9	0.18	0	
	≥$650,000	0.42	0.71	0.24	0.57	0	0	
Smoking status	Never smoker	51.53	50.02	52.43	49.35	58.47	50.89	0.0009
	Past smoker	35	31.51	34.5	40.18	32.05	41.13	
	Current smoker	13.47	18.47	13.07	10.47	9.48	7.97	
Alcohol consumption	Non-drinker	48.98	44.81	48.64	51.99	55.8	49.66	<0.0001
	Former drinker	21.23	19.85	19.37	21.42	23.2	34.92	

	1–2 days/month	18.45	18.89	20.11	17.43	18.41	10.64	
	≥3 days/month	11.34	16.45	11.88	9.15	2.59	4.78	
Physical activity	Inactive	16.18	12.79	14.37	18.69	20.82	26.81	0.0002
	Light/moderate activity	22.25	19.25	21.84	23.72	29	23.31	
	Moderate/vigorous activity	61.57	67.96	63.79	57.59	50.18	49.87	

Abbreviations: BMI = Body mass index; HRS = Health and Retirement Study; SE = Standard error.

Note: Body mass index (BMI) was categorized as normal weight (18.5 to <25 kg/m^2), overweight (25 to <30 kg/m^2), obesity class I (30 to <35 kg/m^2), obesity class II (35 to <40 kg/m^2), and obesity class III (≥40 kg/m^2).

[a] Descriptives are based on the women-only HRS analytic sample (n=2,885) aged ≥50 years in 2006 with complete data. Estimates account for the complex survey design using 2006 Enhanced Face-to-Fact Interview weights and Taylor linearization. Continuous variables are presented as survey-weighted means and standard errors, and categorical variables as survey-weighted row percentages. Age was categorized as 50–59, 60–69, and ≥70 years. Covariates include race/ethnicity, marital status, education, household wealth, smoking, alcohol use, and physical activity.

Table S.3. Incident type 2 diabetic cases, person-years of follow-up, and incidence rates of type 2 diabetes by body mass index categories and body mass index-specific waist circumference categories in the HRS analytic sample (n=2,885) [a]

Category	Cases/Sample	Person-Years	Rate (95% confidence interval) [per 10,000 person-years]
Body mass index:			
Normal	42 / 857	12505.53	33.59 (24.21, 45.4)
Overweight	101 / 955	13422.51	75.25 (61.29, 91.43)
Obesity I	123 / 631	8348.26	147.34 (122.45, 175.79)
Obesity II	51 / 275	3648.12	139.8 (104.09, 183.81)
Obesity III	39 / 167	2126.07	183.44 (130.44, 250.76)
Body mass index-specific waist circumference categories:			
Normal, WC <80	12 / 318	4682.20	25.63 (13.24, 44.77)
Normal, WC ≥80	30 / 539	7823.33	38.35 (25.87, 54.74)
Overweight, WC <90	29 / 298	4233.17	68.51 (45.88, 98.39)
Overweight, WC ≥90	72 / 657	9189.34	78.35 (61.31, 98.67)
Obesity I, WC <105	66 / 376	5046.16	130.79 (101.15, 166.4)
Obesity I, WC ≥105	57 / 255	3302.09	172.62 (130.74, 223.65)
Obesity II, WC <115	30 / 172	2304.07	130.2 (87.85, 185.87)
Obesity II, WC ≥115	21 / 103	1344.04	156.24 (96.72, 238.84)
Obesity III, WC <115	9 / 48	631.02	142.63 (65.22, 270.75)
Obesity III, WC ≥115	30 / 119	1495.04	200.66 (135.39, 286.46)
Overall	356 /2,885	40,050	88.89(79.97,98.15)

Abbreviations: BMI = Body mass index; HRS = Health and Retirement Study. *Note:* Body mass index (BMI) was categorized as normal weight (18.5 to <25 kg/m^2), overweight (25 to <30 kg/m^2), obesity class I (30 to <35 kg/m^2), obesity class II (35 to <40 kg/m^2), and obesity class III (≥40 kg/m^2).

[a] Incident diabetes was identified among women ≥ 50 years free of diabetes at the 2006 baseline and followed through 2020. Follow-up accrued until type 2 diabetes onset, death, loss to follow-up, or administrative censoring (January 1, 2021). Incidence rates (per 10,000 person-years) were estimated using the exact Poisson method with 95% confidence intervals. BMI was categorized using standard cut points, and waist circumference (WC), measured during the Enhanced Face-to-Face assessment, was standardized to centimeters and classified using BMI-specific thresholds (≥80, ≥90, ≥105, and ≥115 cm). Analyses accounted for the complex survey design using sampling weights, strata, and primary sampling units.

Table S.4. Descriptive characteristics of HANDLS dataset by exposure variables (n=1,037) [a]

Characteristic	Category	Overall (n=1037)	Normal (n=241)	Overweight (n=284)	Obesity I (n=223)	Obesity II (n=152)	Obesity III (n=137)	P value
N		1037	241	284	223	152	137	
Age	Mean ± SD	47.69 ± 9.24	46.55 ± 9.01	48.72 ± 9.20	48.53 ± 9.40	46.35 ± 9.34	47.70 ± 9.39	0.016
BMI	Mean ± SD	31.10 ± 1.89	22.17 ± 1.73	27.37 ± 1.41	32.34 ± 1.44	37.31 ± 1.43	45.64 ± 4.42	<0.0001
WC	Mean ± SD	99.66 ± 8.25	80.51 ± 7.11	93.42 ± 7.36	103.28 ± 8.28	111.82 ± 9.57	126.93 ± 10.61	<0.0001
HEI-2010	Mean ± SD	43.51 ± 11.93	45.07 ± 13.53	42.69 ± 12.27	44.20 ± 11.67	41.00 ± 10.14	44.11 ± 10.84	0.040
Follow-up time from baseline	Mean ± SD	7.96 ± 2.15	8.35 ± 2.31	7.96 ± 2.20	7.82 ± 2.06	7.96 ± 2.06	7.50 ± 2.04	0.0047
Race category	White	40.7	46.9	42.6	35.9	39.5	35	0.081
	African American	59.3	53.1	57.4	64.1	60.5	65	
Baseline age category	<50	58.43	64.7	52.8	55.6	65.8	55.5	0.045
	50-59	30.1	25.7	33.5	30	25.7	35.8	
	60-69	11.46	9.5	13.7	14.3	8.6	8.8	
Education category	Less than high school	30.56	31.5	23.9	35.2	32.7	32.8	0.0038
	High school	34.19	27.2	38.4	34.2	39.3	32.1	
	Some college	22.67	21.7	24.6	22.8	18.7	24.6	
	College+	12.53	19.6	13	7.8	9.3	10.4	
Poverty status category	Above	57.86	51.9	63	58.3	52.6	62.8	0.045
	Below	42.14	48.1	37	41.7	47.4	37.2	
Smoking category	Never tried	26.04	24.3	27	25.9	25	28.5	0.00051
	Tried, never used regularly	10.91	8.4	10.3	10	16.2	12.2	
	Former user	19.18	9.8	19	24.4	21.3	25.2	
	Current user	43.89	57.5	43.7	39.8	37.5	34.1	
Alcohol category	Never tried	10.89	7.9	12.5	9	10.4	16.4	0.098
	Tried, never used regularly	9.6	7.9	9.5	10.9	10.4	9.8	
	Used >6 months ago	24.63	20.6	22	27.9	31.9	23.8	

	Used in past 6 months	54.92	63.6	56.1	52.2	47.4	50	
High waist circumference using 80 cm threshold	WC <80 cm	11.79	46.5	3.2	0	0.7	0	<0.0001
	WC ≥80 cm	88.21	53.5	96.8	100	99.3	100	
BMI-specific high waist circumference	Lower WC within BMI	42.63	46.5	28.2	58.7	63.8	16.1	<0.0001
	Higher WC within BMI	57.37	53.5	71.8	41.3	36.2	83.9	
Incident diabetes at HANDLS follow-up	No	86.14	96.7	89.8	82.1	80.3	73	<0.0001
	Yes	13.86	3.3	10.2	17.9	19.7	27	

Abbreviations: BMI = Body mass index; HANDLS = Healthy Aging in Neighborhoods of Diversity across the Life Span; HEI-2010 = Healthy Eating Index, 2010 version; SD = Standard deviation; WC = Waist circumference.

[a] Values are presented as mean ± standard deviation (SD) for continuous variables and n (%) for categorical variables. P-values were derived using analysis of variance (ANOVA) for continuous variables and χ^2 tests for categorical variables. BMI categories were defined as normal weight (18.5–24.9 kg/m²), overweight (25.0–29.9 kg/m²), obesity class I (30.0–34.9 kg/m²), obesity class II (35.0–39.9 kg/m²), and obesity class III (≥40.0 kg/m²). Descriptive statistics were based on the observed m=0 MI-eligible analytic dataset.

Table S.5. Incident type 2 diabetic cases, person-years of follow-up, and incidence rates of type 2 diabetes by body mass index categories and body mass index-specific waist circumference categories in the HANDLS analytic sample (n=1,037) [a]

Category	Cases/Sample	Person-Years	Rate (95% confidence interval) [per 10,000 person-years]
Body mass index:			
Normal	8/241	2012.5	39.8(17.2,78.3)
Overweight	29/284	2259.4	128.4(86.0,184.3)
Obesity I	40/223	1742.9	229.5(167.0,312.5)
Obesity II	30/152	1210.6	247.8(167.2,353.8)
Obesity III	37/137	1027.9	360.0(253.4,496.2)
Body mass index-specific waist circumference categories:			
Normal, WC <80	4/112	952.8	42.0 (11.4,107.5)
Normal, WC ≥80	4/129	1060	37.7 (10.3,96.6)
Overweight, WC <90	7/80	704	99.4(40.0,204.8)
Overweight, WC ≥90	22/204	1555	141.5(88.7,214.2)
Obesity I, WC <105	19/131	1063	178.7(107.6,279.0)
Obesity I, WC ≥105	21/92	680	309.0(191.3,472.3)
Obesity II, WC <115	16/97	776	206.1(117.8,334.7)
Obesity II, WC ≥115	14/55	434	322.3(176.2,540.7)
Obesity III, WC <115	5/22	197	253.4(82.2,591.4)
Obesity III, WC ≥115	32/115	831	385.3(263.5,543.9)
Overall	144/1,037	8,253	174.5(147.1,205.7)

Abbreviations: BMI = Body mass index; HANDLS = Healthy Aging in Neighborhoods of Diversity across the Life Span; WC = Waist circumference. *Note:* Body mass index (BMI) was categorized as normal weight (18.5 to <25 kg/m^2), overweight (25 to <30 kg/m^2), obesity class I (30 to <35 kg/m^2), obesity class II (35 to <40 kg/m^2), and obesity class III (≥40 kg/m^2).

[a] Rates are expressed per 10,000 person-years. Incident type 2 diabetes was defined as new-onset diabetes after baseline among participants free of prevalent diabetes at Visit 1. Person-years were accrued until diabetes onset, censoring, death, or end of follow-up. Confidence intervals were based on exact Poisson methods. BMI categories were defined using standard cut-points, while BMI-specific WC categories used thresholds of 80 cm (normal weight), 90 cm (overweight), 105 cm (obesity class I), and 115 cm (obesity classes II–III). Estimates were based on the observed m=0 MI-eligible analytic dataset and are presented descriptively without Rubin pooling.